Cryptography in ancient India

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Abstract - Cryptography has its own advantages now days for hiding information. As day-by-day uses of internet are increasing there is need to transfer information in unknown format. But it is not only now days we are using for information transfer, it has tales in past for hiding information in mythological text for private messaging or changing the meaning of text. The paper throws light on techniques like Katapayadi, Ramshalaka and Bhutasamkhya and other system of cryptography in ancient India.

Keywords: Information security, Encryption, Decryption, Cryptography

Introduction

Cryptography is procedure of conversion of text from one form to another to change original meaning. It has two stages, first is encryption is conversion of plain text to unknown format and decryption is conversion-encrypted format to original text. The unknown format after encryption is known as cipher text. There are two types of cryptography first is substitution cryptography in which letter of plain text are substituted with other letter to change plain text meaning to unknown and vice-versa to get original text. Substitution method also has two types, first is monographic substation where single letter is replace for conversion. In polygraphic method group of letters are used to change the meaning of original message. Another way of cryptography is transposition in which rearrangement of letters is use to change the meaning of original text.

In ancient India, cryptography was used in Indian mythological text like hymns and lyrics for hiding text. Some of the methods of were as follows

KaTaPaYaDi Sankhya

KaTaPaYaDi [1] is used in Sanskrit texts as a hashing technique. Each consonant of an Indian language is assign a number in this Katapayadi system as follows:

Table 1

कटपयादि संख्या - kaTapayAdi for Melakarta Ragam Names & Numbers									
1	2	3	4	5	6	7	8	9	0
क	ख	ग	घ	ङ	च	छ	ज	झ	ञ
ट	ठ	ड	ढ	ण	त	थ	द	ध	न
प	फ	ब	भ	म					
य	र	ल	व	श	ष	स	ह		
ka	kha	ga	gha	nga	cha	Cha	ja	Jha	nya
Ta	Tha	Da	Dha	Na	ta	tha	da	dha	na
pa	pha	ba	bha	ma					
ya	ra	la	va	sha	Sha	sa	ha		

Suppose we want to hash the word 'Bharat', the number based on katapayadi system would be then as follows for 'Bharat'

Bha= Bha = 4
Ra = Ra = 2
T = ta = 6

Therefore, number 624 represents Bharat

Indian hashes are reversed numbers that is why written 624 and Not 426.

Ramshalaka

Ramshalaka is transposition substitution cryptography. As shown in column 1 of table 2 “Ramshalaka uses nine stanzas or chopai of Philosophical meaning, from which solution to any actions/questions related to our activities and lives are incidental and in-built, in a square matrix of size 15x15” [1]. Shalaka means matrix, and this shalaka was use by Ram who was famous king and god in Indian mythology ‘Ramayana’. Ram used this technique to find answers questions of that is why it called as Ramshalaka.

Verses generally called stanza or rhymes are in first column of Table 3. These verses are broken into independent ‘akshar’ (Hindi characters) as shown in third column of Table 3 and these all akshars creates 15x15 matrix, which is actual shalaka. It is difficult to find or recognize verses from table as that are in matrix form. This means actual text can kept hidden in Shalaka and all the hidden stanzas can be decrypted using some key. In shalaka, order of characters of original text was changed. From Table 4 and Table 5 encrypted technique is visible. While encrypting first character is taken from each, of nine chopais after first character finishes form all stanzas, second character from each chopai is taken and similarly this procedure is followed to form 15x15 matrix.

Table 2 RamShalaka

सु	प्र	उ	बि	हो	मु	ग	व	सु	नु	बि	घ	धि	इ	द
र	रु	फ	सि	सि	रें	बस	है	मं	ल	न	ल	य	न	अं
सुज	सो	ग	सु	कु	म	स	ग	त	न	ई	ल	धा	बे	नो
त्य	र	न	कु	जो	म	रि	र	र	अ	की	हो	सं	रा	य
पु	सु	थ	सी	जे	इ	ग	म	सं	क	रे	हो	स	स	नि
त	र	त	र	स	इ	ह	ब	ब	प	चि	स	य	स	तु
म	का	ा	र	र	मा	मि	मी	म्हा	ा	जा	हू	हीं	ा	जू
ता	रा	रे	री	हृ	का	फ	खा	जि	ई	र	रा	पू	द	ल
नि	को	मि	गो	न	म	ज	य	ने	मनि	क	ज	प	स	ल
हि	रा	म	स	रि	ग	द	न	ष	म	खि	जि	मनि	त	जं
सिं	मु	न	न	कौ	मि	ज	र	ग	धु	ख	सु	का	स	र
गु	क	म	अ	ध	नि	म	ल	ा	न	ब	ती	न	रि	भ
ना	पु	ब	अ	ढा	र	ल	का	ए	तु	र	न	नु	ब	थ
सि	ह	सु	म्हा	रा	र	स	हिं	र	त	न	ष	ा	जा	ा
र	सा	ा	ला	धी	ा	री	ज	हू	हीं	खा	जू	ई	रा	रे

Table 3 Ramshalaka characterization

9 VERSES/COUPLET(*CHOPAI*)	No. of Letters	*25 AKSHARs* used in *Shalaka*
1. ***सुनु सिय सत्य असीस हमारी पूजिहि मन कामना तुम्हारी*** *(Sun siya satya asis hamari, Pujahi mankaamna tumahri)*	23	सु नु सि य स त्य अ सी स ह मा री पू जि हि म न का म ना तु म्ह ा री
2. ***प्रबिसि नगर कीजे सब काजा हृदयँ राखि कोसल पुर राजा*** (*Prabasi nagar keeje sab kaaja, Hirde rakhi kausalpur raja*)	*24*	प्र बि सि न ग र की जे स ब का जा ह्र दयँ रा खि को स ल पु र रा ज ा
3. ***उधरहिं अंत न होइ निबाहू काल नेमि जिमि रावन राहू*** (Udhre ant na hoi nibahu, kalnemi jimi ravan rahu)	*23*	उ ध रहिं अं त न हो इ नि बा ह् का ल ने मि जि मि रा व न रा हू
4. ***बिधि बस सुजन कुसंगत परहीं फनि मनि सम निज गुन अनुसरहीं*** (*Udhre ant na hoi nibahu, kalnemi jimi ravan rahu)*	*29*	बि धि बस सुज न कु सं ग त प र हीं फ नि मनि स म निज गु न अ नु स र हीं
5. ***होइहि सोइ जो राम रचि राखा कोकरि तर्क बढ़ावै साखा*** (*Hoi hai soi jo Ram rachi rakha, Ko kari tarak badavahi saka)*	*22*	हो इ हि सो इ जो रा म र चि रा खा को क रि त र्क ब ढ़ा वै सा खा
6. ***मुद मंगलमय संत समाजू जो जग जंगम तीरथ राजू*** *(Mud mangalmay sant samaju, jimi jag langam teerath raju)*	*23*	मु द मं ग ल म य सं त स मा जू जो ज ग जं ग म ती र थ रा जू
7. ***गरल सुधा रिपु करहीं मिताई गोपद सिंधु अनल सितलाई*** (*Garal sudha ripu karay mitai, Gopad sindhu anal sitlai)*	*25*	ग र ल सु धा रि पु क र हीं मि ता ई गो प द सिं धु अ न ल सि त ला ई
8. ***बरुन कुबेर सुरेस समीरा रन सन्मुख धरि काहुँ न धीरा*** *(Varun kuber sures samira, ran sammukh dhar kah na dheera)*	*24*	ब रू न कु बे र सु रे स स मी रा र न स न्मु ख ध रि का हुँ न धी रा
9. ***सुफल मनोरथ होहुँ तुम्हारे रामु लखनु सुनि भए सुखारे*** *(Sufal manorath hoy tumhare, Ram Lakhan suni bhaye sakhare)*	*27*	सु फ ल म नो र थ हो हुँ तु म्हा रे रा मु ल ख नु सु नि भ ए सु ख ा रे

Table 4

सु	प्र	उ	बि	हो	मु	ग	ब	सु	नु	बि	घ	धि	इ	द
र	रु	फ	सि	सि	रहिं	बस	हि	मं	ल	न	ल	य	न	अं
सुज	सो	ग	सु	कु	म	स	ग	त	न	इ	ल	धा	बे	नो
त्य	र	न	कु	जो	म	रि	र	र	अ	की	हो	सं	रा	य
पु	सु	थ	सी	जे	इ	ग	म	सं	क	रे	हो	स	स	नि
त	र	त	र	स	हुँ	ह	ब	ब	प	चि	स	हिं	स	तु
म	का	ा	र	र	म	मि	मी	म्हा	ा	जा	हू	हीं	ा	ा
ता	रा	रे	री	ह	का	फ	खा	जू	ई	र	रा	पू	द	ल
नि	को	जो	गो	न	मु	ज	य	ने	मनि	क	ज	प	स	ल
हि	रा	मि	स	रि	ग	द	न्मु	ख	म	खि	जि	म	त	जं
सि	ख	नु	न	को	मि	निज	कं	ग	धु	ध	सु	का	स	र
गु	ब	म	अ	रि	नि	म	ल	ा	न	ढ़	ती	न	क	भ
ना	पु	व	अ	ा	र	ल	ा	ए	तु	र	न	नु	वै	थ
सि	हु	सु	म्हा	रा	र	स	स	र	त	न	ख	ा	ज	ा
र	ा	ा	ला	धी	ा	री	ा	हू	हीं	खा	जू	ई	रा	रे

Table 5

सु	प्र	उ	बि	हो	मु	ग	ब	सु	नु	बि	घ	धि	इ	द
र	रु	फ	सि	सि	रहिं	बस	हि	मं	ल	न	ल	य	न	अं
सुज	सो	ग	सु	कु	म	स	ग	त	न	इ	ल	धा	बे	नो
त्य	र	न	कु	जो	म	रि	र	र	अ	की	हो	सं	रा	य
पु	सु	थ	सी	जे	इ	ग	म	सं	क	रे	हो	स	स	नि
त	र	त	र	स	हुँ	ह	ब	ब	प	चि	स	हिं	स	तु
म	का	ा	र	र	म	मि	मी	म्हा	ा	जा	हू	हीं	ा	ा
ता	रा	रे	री	ह	का	फ	खा	जू	ई	र	रा	पू	द	ल
नि	को	जो	गो	न	मु	ज	य	ने	मनि	क	ज	प	स	ल
हि	रा	मि	स	रि	ग	द	न्मु	ख	म	खि	जि	म	त	जं
सि	ख	नु	न	को	मि	निज	कं	ग	धु	ध	सु	का	स	र
गु	ब	म	अ	रि	नि	म	ल	ा	न	ढ़	ती	न	क	भ
ना	पु	व	अ	ा	र	ल	ा	ए	तु	र	न	नु	वै	थ
सि	हु	सु	म्हा	रा	र	स	स	र	त	न	ख	ा	ज	ा
र	ा	ा	ला	धी	ा	री	ा	हू	हीं	खा	जू	ई	रा	रे

Bhutasamkhya system

Bhutasamkhya system is a method of representing numbers using words having meanings of numerical values. Bhutasamkhya system words chosen from Sanskrit language from which words chosen to replace for numbers. The number 2 represented with the word "eye" as every human being has two eyes. Similarly, number “32” represented by word “Teeth”.

Small numbers represented by single words. Where sentences used for representing large numbers. To decode numbers from sentences, digits written from left to right. Varahamihira [2](505 – 587 CE), an Indian astronomer, mathematician, and astrologer, encoded this number as “kha kha aṣṭi yamaḥ”. The individual words in this are "*kha*", "*kha*", "*aṣṭi*" and "*yamāḥ*" and they denote numbers "0", "0", "61" and "2" respectively. To obtain number indicated by kha-kh-aṣṭi-yamaḥ, the numbers need to arrange in the reverse order, as "2", "16","0","0". Placing numbers side by side, we get the number 21600.

Kautilayama

“Kautilayama [1] in which letters were substituted based on phonetic relation” and other is there had been development of another version of private messaging such as methods based on knowledge of vernaculars and messaging by wrist-finger gesture known as akshar mustika kathanam.

Muladivaya

“Muladivaya is technique in which paired letters were used as cryptographic unit.” [3]

Aryabhata’s Substitution Code

Aryabhata’s Substitution Code [4] is a plotting of numbers to words that divided into two groups of consonants and vowel.

First 25 consonants called varga letters of Sanskrit alphabet are:

k=1	kh=2	g=3	gh=4	ṅ=5
c=6	ch=7	j=8	jh=9	ñ=10
ṭ=11	ṭh=12	ḍ=13	ḍh=14	ṇ=15
t=16	th=17	d=18	dh=19	n=20
p=21	ph=22	b=23	bh=24	m=25

Other remaining 8 consonants are avarga letters:

y=30	r=40	l=50	v=60
ś=70	ṣ=80	s=90	h=100

The varga letters (k to m) represents squares such as 1, 100, 10000,
The avarga letters (y to h) represents non- squares such as 10, 1000,

This creates a notational system in place values for numbers as large as 10^{17}:

17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
au	au	ai	ai	o	o	e	e	ḷ	ḷ	ṛ	ṛ	u	u	i	i	a	a
औ	औ	ऐ	ऐ	ओ	ओ	ए	ए	ऌ	ऌ	ऋ	ऋ	उ	उ	इ	इ	अ	अ

A letter is to be insert next to its vowel.
For example: if number 4652 is

i	i	a	a
4	6	5	2
gh	c	n	kh

Therefore, number 4652 will; be represented by 4652 = ghicinikhi

Gudhayojya

Gudhayojya [5] was a basic method of changing meaning of true content of spoken messages. In this method of cryptography, unnecessary letters added at beginning or at end of each word, so its original meaning was changed. For example to change meaning of statement “I truly believe in lord Ganesha” on may add letter “ish” at beginning of each word, So sentence become “ishI ishtruly ishbelieve ishin ishlord ishGanesha”, The real content of sentence got changed.

References

[1] R. Prasad, "SRI RAMSHALAKA: A VEDIC METHOD OF TEXT ENCRYPTION AND DECRYPTION," *Indian Journal of Computer Science and Engineering,* vol. 4, no. 3, p. 10, 2013.

[2] "www.britannica.com," Encyclopaedia Britannica, Inc., [Online]. Available: http://www.britannica.com/biography/Varahamihira. [Accessed 05 10 2015].

[3] S. Kumar and R. Prasad, "Some Ancient And Modern Concepts Of Cryptography," *Ideal Science Review,* vol. 4, no. 1, pp. 15-20, 10 2013.

[4] S. Kak, "Aryabhata’s Mathematics," in *RSA Conference*, 2006.

[5] A. B. Ganguly, Fine Arts in Ancient India, Abhinav Publications, p. 170.

[6] A. Raman, "The Ancient Katapayadi Formula And The Modern Hashing Method," p. 10.

[7] A. Raman, "http://citeseerx.ist.psu.edu," [Online]. Available: http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.18.9659&rep=rep1&type=pdf. [Accessed 23 09 2015].